Matrix functions and stability

A pointwise Lipschitz lower bound for the joint spectral radius

Difficulty: challenging **Importance:** interesting to the community

Status: Solved

Rating rationale: Challenging because the sharp one-sided perturbation rate must survive reducibility; community impact is reliable lower stability estimates under data perturbation.

Affirmative resolution - 2026-09-12

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The full pointwise lower-Lipschitz assertion holds. The complete proof, Theorem (1) and Sections 3-6, covers every fixed nonempty compact complex matrix family in every finite dimension, including reducible families with unbounded normalized products. For that fixed reference there are constants $C, r > 0$ such that every nonempty compact family within Hausdorff distance r satisfies the original lower bound below.

The proof establishes a product-bounded exterior-power family at a maximal critical degree and transfers a robust one-sided estimate back to the original family. The nearby family may be arbitrary; it need not preserve the reference's invariant subspaces. Proof PDF · Standalone proof TeX.

The complete clarified proof passed a separate independent Codex-agent mathematical audit. The submission record contains the exact sources, source clarification, supplementary checks, document inspection and bounded public-network/source audit. Substantial AI assistance is disclosed; this is informal agent review, not external human peer review or formal verification.

Epperlein and Wirth retain attribution for the target. The Barabanov/Wirth extremal-norm theorem and Chitour-Mason-Sigalotti nonresonance theory retain credit; the proof supplies its needed paired-tensor arguments in full. The result is pointwise at a fixed family and does not claim a two-sided Lipschitz estimate uniform over two varying families. The original definitions, statement, historical partial status evidence and ratings below are retained.

Context and notation

Let $\mathcal{H}_d$ denote the nonempty compact subsets of $\mathbb{C}^{d\times d}$. Use the spectral norm and its Hausdorff distance

$$d_H(\mathcal{M},\mathcal{N}) = \max\left\{\sup_{A\in\mathcal{M}}\inf_{B\in\mathcal{N}}\|A-B\|_2, \sup_{B\in\mathcal{N}}\inf_{A\in\mathcal{M}}\|A-B\|_2\right\}.$$

The joint spectral radius is

$$\hat{\rho}(\mathcal{M}) = \lim_{k\to\infty}\max_{A_1,\dots,A_k\in\mathcal{M}}\|A_k\cdots A_1\|_2^{1/k}.$$

These definitions also apply to finite real matrix sets. The ordinary spectral radius of one matrix is written $\rho(A)$.

Problem statement

For every $d \geq 1$ and $\mathcal{M} \in \mathcal{H}_d$, do there exist $r, C > 0$ such that

$$\hat{\rho}(\mathcal{N}) \geq \hat{\rho}(\mathcal{M}) - C d_H(\mathcal{M},\mathcal{N}) \quad \text{if } d_H(\mathcal{M},\mathcal{N}) < r?$$

Reference and status evidence

Epperlein and Wirth, The joint spectral radius is pointwise Hölder continuous, §2, Conjecture 3 (P2). The conjectured exponent is one.

Scope

The reference family is fixed while the perturbed family varies. An answer controls how rapidly a stability diagnostic can decrease under data error; MF-05 instead requires a two-sided estimate uniform over two varying families.

Additional status evidence

Searches combining "joint spectral radius" with "local Hölder", "Lipschitz lower", "trajectory bounds", the authors' names, and 2025/2026 found no later resolution. These searches supplement the explicit 2025 conjectures; they do not establish exhaustiveness. The three entries are separately named assertions in the source, not a count of dimensional cases.

Audit — 2026-09-10

Rechecked Epperlein–Wirth, §1 and Conjecture 3 (P2). Local Lipschitz continuity gives the asserted lower bound at irreducible families. The general reducible case remains posed, and searches for Lipschitz lower bounds and later author work found no resolution.